

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 11A.6

EFFECT OF TEMPERATURE ON THE RATE CONSTANT

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A.6: Effect of Temperature on the Rate Constant
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	10 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 11A.6

Q1. Q1. State the Arrhenius equation and define all terms within it.

[4 Marks]

WCH14/01/Jan23/Q5(a)

Q2. Q2. A graph of $\ln k$ against $1/T$ is plotted for a reaction. The gradient is -10500 K . Calculate the activation energy in kJ mol^{-1} . ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3 Marks]

WCH14/01/Jan23/Q5(b)

Q3. Q3. What does the y-intercept of a graph of $\ln k$ against $1/T$ represent?

[1 Marks]

WCH14/01/Oct21/Q6(a)

Q4. Q4. Explain the effect of increasing temperature on the value of the rate constant, k .

[2 Marks]

WCH14/01/Oct21/Q6(b)

SECTION B: WORKED MARK SCHEME — 11A.6

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Q1. State the Arrhenius equation and define all terms within it. • $k = A e^{(-E_a/RT)}$ (1) • k = rate constant, A = pre-exponential/frequency factor (1) • E_a = activation energy, R = gas constant (1) • T = temperature in Kelvin (1)	[4]
Q2	Q2: Q2. A graph of $\ln k$ against $1/T$ is plotted for a reaction. The gradient is -10500 K . Calculate the activation energy in kJ mol^{-1} . ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) • $\text{gradient} = -E_a / R$ (1) • $-10500 = -E_a / 8.31$ (1) • $E_a = +87255 \text{ J mol}^{-1} = +87.3 \text{ kJ mol}^{-1}$ (1)	[3]
Q3	Q3: Q3. What does the y -intercept of a graph of $\ln k$ against $1/T$ represent? • $\ln A$ (the natural log of the pre-exponential factor) (1)	[1]
Q4	Q4: Q4. Explain the effect of increasing temperature on the value of the rate constant, k . • k increases exponentially as temperature increases (1) • Because the term $e^{(-E_a/RT)}$ increases (1)	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 11A.6**FAQ #1: What are the units for activation energy?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What are the units for activation energy?

MODEL ANSWER / EXAMINER EXPECTATION:

When calculated using $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$, E_a is in J mol^{-1} . It is usually converted to kJ mol^{-1} for the final answer.

FAQ #2: Do I need to convert temperature to Kelvin?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Do I need to convert temperature to Kelvin?

MODEL ANSWER / EXAMINER EXPECTATION:

Yes, T must always be in Kelvin (K) when using the Arrhenius equation.

FAQ #3: What is the significance of the pre-exponential factor A ?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is the significance of the pre-exponential factor A ?

MODEL ANSWER / EXAMINER EXPECTATION:

It represents the frequency of collisions with the correct orientation for reaction.